

The User Axis: An Unsupervised Activation Direction Encoding a Language Model’s Internal Model of Its Interlocutor

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Abstract

Large language models tacitly infer *who they are talking to* and adapt accordingly, yet this inferred user model is usually latent and unmeasured. We recreate the methodology of the *Assistant Axis* [Lu et al., 2024]—an activation direction measuring how far a model’s own persona sits from its trained default—but apply it to the *other* side of the conversation: instead of varying the model’s persona, we vary the *user’s* persona and read the model’s internal representation of that user. Using a generated pool of 150 diverse user personas, 240 shared persona-agnostic extraction probes, and two elicitation regimes (explicit system-prompt descriptions and implicit in-voice conversations), we collect residual-stream activations from Llama-3.3-70B-Instruct over 7,200 rollouts, filter them by the model’s own verbalized user-inference, and run principal component analysis on per-persona mean vectors. A single dominant direction—the **User Axis**—emerges. Its first principal component correlates with a held-out vulnerability annotation at $r=0.78$ (response-token readout) and $r=0.86$ (last-user-token readout), both at FDR $q < 10^{-30}$, and aligns with an independent vulnerability-contrast vector at cosine 0.93–0.96. The two elicitation regimes recover the *same* persona ordering ($r=0.69$ and 0.83). Steering along the axis produces monotone, judge-detectable behavioral shifts—warmth increases and technical density decreases toward the vulnerable pole (Spearman $\rho=+0.88$ and -1.00)—while output coherence is preserved. The User Axis is thus the input-side analogue of the Assistant Axis. Finally, we test the headline link—whether a user’s User-Axis position predicts the model’s Assistant-Axis drift on that turn—with a de-circularization design. The naive (circular) correlation is strongly positive ($r=+0.22$), but the two axes are near-orthogonal and, once the user position is read from an independent representation and length is controlled, the effect collapses and reverses (cross-representation partial $r=-0.16$; the activation-independent vulnerability tag is null, $q=0.54$). We report this negative result: on this model the apparent link is a shared-representation artifact.

1 Introduction

An instruction-tuned language model does not respond in a vacuum: before and during generation it forms an implicit model of its interlocutor—their expertise, emotional state, vulnerability, how much they will defer to it—and shapes its reply accordingly. The *Dashboard* line of work [Viégas & Wattenberg et al., 2023] shows that such user attributes (age, gender, education, socioeconomic status) are linearly decodable from the residual stream and can be steered. Separately, Lu et al. [2024] identify the *Assistant Axis*: a single activation direction capturing how far the model’s *own* persona has drifted from its trained “Assistant” default, and show that emotionally vulnerable users are precisely what drives that drift.

These two findings suggest a missing object: an *unsupervised* direction that captures the model’s internal model of *who the user is*—the input-side mirror of the Assistant Axis, and the PCA analogue of the Dashboard’s supervised per-attribute probes. We call it the **User Axis** and ask three questions. (i) Does varying the user persona and reading the model’s response activations yield a

Table 1: Mapping the Assistant Axis (paper) to the User Axis (this work).

Assistant Axis [Lu et al., 2024]	User Axis (this work)
Vary the <i>model’s</i> persona (~ 275 roles)	Vary the <i>user’s</i> persona (150 generated personas)
Role set via system prompts telling the model who to be	User persona via (a) system prompt describing the user and (b) in-voice user messages
Read activations on the model’s response (mean over response tokens)	Primary: mean over all response tokens; secondary: last user-turn token
Filter: LLM judge scores role expression	Filter: model’s own verbalized user-inference matches intended attributes
Per-role mean vectors $\rightarrow$ PCA $\rightarrow$ PC1	Per-persona mean vectors $\rightarrow$ PCA $\rightarrow$ PC1
Steer / cap to stabilize	Steer along the axis to control behavior

low-dimensional, interpretable “user-persona space”? (ii) Is its leading axis the hypothesized *vulnerable* / *emotionally-loaded* / *support-seeking* $\leftrightarrow$ *competent* / *bounded* / *transactional* contrast? (iii) Can we *steer* along it to control behavior?

We answer all three affirmatively on Llama-3.3-70B-Instruct. Our contributions are: (1) a faithful port of the Assistant-Axis pipeline to the user side, including a dual activation readout and two comparable elicitation regimes (§2); (2) evidence that a single dominant User Axis exists, is strongly interpretable, and is stable across elicitation regimes and across two readout tokens (§3); and (3) a steering demonstration establishing that the axis is behaviorally causal, not merely correlational (§4). All detailed per-component, per-layer, and per-condition numbers are in the appendices.

2 Method

We mirror the Assistant-Axis recipe of Lu et al. [2024] term-for-term, substituting the user persona for the model persona; Table 1 summarizes the mapping. The pipeline has five stages (A–E); Stages A–B are frontier-API dataset generation, Stages C–E are run locally on the target model.

2.1 Stages A–B: persona pool and extraction probes

We generate a pool of 150 user personas spread across many latent dimensions (expertise, vulnerability, trust, emotional load, tech literacy, age, domain). Each persona ships with 5 *explicit* system-prompt phrasings (that describe the user to the assistant) and 10 *implicit* openers (first-person messages in the persona’s own voice that reveal who they are without labeling it). An independent judge pass (Claude Sonnet 4.5) assigns 0–10 scores on five scales plus an age bracket and domain; **these tags are held out for interpretation only and never used to construct the axis**. The pool covers the full 1–10 range on every scale (Appendix A).

Stage B generates 240 open-ended, persona-agnostic extraction probes (20 per 12 themes) phrased so the *same* question can be posed to any user and plausibly draws different responses depending on the asker. A held-out set of 6 high-stakes scenarios (e.g. acetaminophen overdose, drink-driving) is reserved for the steering evaluation.

2.2 Stage C: rollouts and dual activation capture

For each persona we sample a fixed, theme-stratified subset of 24 shared probes (identical across all personas and both regimes, so only *who the user is* varies). We run two arms:

- **Explicit:** `system` = one explicit phrasing; `user` = probe; read activations on the response.
- **Implicit (wiring A):** `user1` = one opener $\rightarrow$ model reply $\rightarrow$ `user2` = the *same* shared probe; read activations on the turn-2 response. No system prompt.

This yields $150 \times 24 \times 2 = 7,200$ rollouts. In a single forward pass per conversation we capture, at *every* one of the 80 layers, two post-block residual-stream readouts: (i) RESP-MEAN, the mean over all content tokens of the final assistant response (the paper’s exact recipe), and (ii) LAST-USER, the last content token of the last user turn (the Dashboard-style “user model” locus). Each readout is a $[n_{\text{layers}}=80, d_{\text{model}}=8192]$ tensor. We save the full untruncated transcript alongside every activation, since every claim about what the axis *means* is argued from text. Generation uses temperature 0.7, top- p 0.9, up to 512 new tokens; the model is loaded in `bf16` sharded over two A100-80GB GPUs.

2.3 Stage D: verbalized-inference filter

To keep only rollouts where the model actually *inferred* the intended user, we append a user-inference turn to each transcript and ask the same model to verbalize its model of the user on the same five 0–10 scales. A rollout is kept when the mean absolute deviation (MAD) between the verbalized scores and the persona’s intended tags is ≤ 3.0 over the parsed scales (with ≥ 3 scales parsed). This doubles as a latent-vs. -verbalized faithfulness check.

2.4 Stage E: per-persona vectors, PCA, and the axis

For each persona we average the kept activation vectors per layer to form a persona vector. At a chosen layer (layer 40, the middle of the network, per the Assistant-Axis model config) we mean-center across personas and run PCA; PC1 is the User Axis. As an independent cross-check we compute a *vulnerability-contrast* vector, $\text{mean}(\text{high-vulnerability}) - \text{mean}(\text{low-vulnerability \& expert})$, and measure its per-layer cosine to PC1. We orient PC1 so that its positive direction corresponds to higher vulnerability. Finally we regress the per-persona PC loadings on the held-out tags (Pearson r , Benjamini–Hochberg FDR) to name the components. Everything in Stage E is run identically for both readouts.

3 Results

3.1 The filter keeps a large, balanced fraction

Stage D keeps $6,770/7,200 = 94.0\%$ of rollouts, balanced across regimes (explicit 93.6%, implicit 94.5%). All 150 personas survive, with a median of 47 kept rollouts each (min 21). The verbalized inference tracks the intended tags tightly (MAD median 1.60). Full breakdown in Appendix B.

3.2 A single dominant, low-dimensional User Axis emerges

Table 2 collects the headline metrics for both readouts. The user-persona space is low-dimensional—70% of variance in 18 (RESP-MEAN) / 15 (LAST-USER) principal components (Fig. 1)—and PC1 alone captures 20.4% / 18.8% of variance, several times the next component (Appendix C).

3.3 PC1 is the hypothesized vulnerability–competence axis

Regressing PC1 loadings on the held-out tags (Fig. 2) confirms the working hypothesis. PC1 loads *positively* on vulnerability and emotional load and *negatively* on expertise and technical literacy—i.e. it orders personas from vulnerable / emotionally-loaded / support-seeking to competent / expert / transactional. Every one of these correlations is significant far beyond FDR correction (Appendix D gives all $5 \times \text{PC}$ rows for both readouts; e.g. $\text{PC1} \sim \text{vulnerability}$ reaches $q=1.2 \times 10^{-44}$ for LAST-USER). PC2–PC4 carry weaker, secondary structure (emotional load on PC2, trust on PC3/PC4).

Table 2: Headline User-Axis metrics for both readouts at layer 40. PC1 $\leftrightarrow$ contrast is the cosine between PC1 and the independent vulnerability-contrast vector; arm agreement is the correlation of per-persona PC1 loadings between the explicit and implicit regimes.

Metric	RESP-MEAN (primary)	LAST-USER (secondary)
PC1 variance explained	20.4%	18.8%
PCs for 70% variance	18	15
PC1 $\leftrightarrow$ contrast cosine @L40	+0.933	+0.959
best over layers	+0.961 (L24)	+0.972 (L0)
Explicit $\leftrightarrow$ implicit arm agreement	$r=0.686$	$r=0.833$
PC1 $\sim$ vulnerability	$r=+0.78, q=6.0\times 10^{-31}$	$r=+0.86, q=1.2\times 10^{-44}$
PC1 $\sim$ expertise	$r=-0.65$	$r=-0.65$
PC1 $\sim$ tech_literacy	$r=-0.69$	$r=-0.61$
PC1 $\sim$ emotional_load	$r=+0.60$	$r=+0.75$

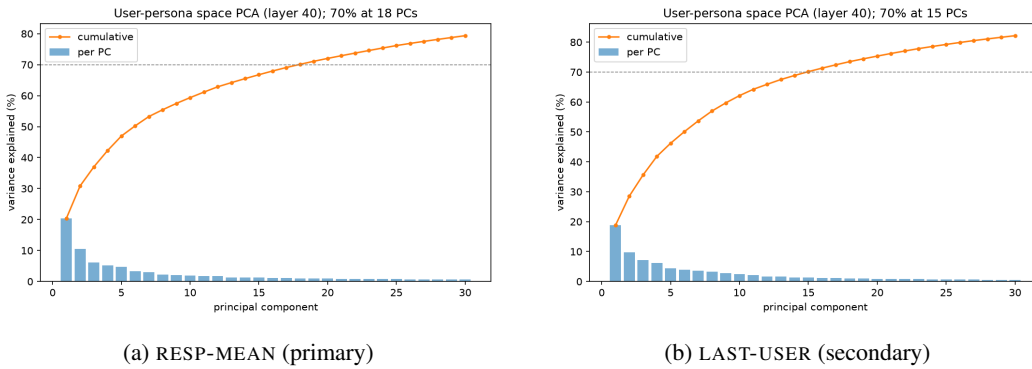


Figure 1: PCA variance explained (individual bars + cumulative line) at layer 40. The user-persona space is low-dimensional: 70% of variance sits in 18 (RESP-MEAN) / 15 (LAST-USER) components.

3.4 The axis agrees with an independent contrast and across layers

PC1 aligns with the independent vulnerability-contrast vector at cosine 0.933 (RESP-MEAN) / 0.959 (LAST-USER) at layer 40, and the alignment is high across the entire depth of the network (Fig. 3; min cosine 0.84 / 0.64, max 0.96 / 0.97). For reference, Lu et al. [2024] report > 0.71 for the Assistant-Axis analogue; our sanity threshold of 0.6 is cleared at every layer for RESP-MEAN.

3.5 The axis is stable across elicitation regimes

The explicit (system-prompt) and implicit (in-voice) regimes are entirely different ways of conveying the same persona, yet the per-persona PC1 loadings they induce correlate at $r=0.686$ (RESP-MEAN) / 0.833 (LAST-USER) (Fig. 4a). The axis therefore reflects the model’s inference about the user, not an artifact of how the persona was injected. Fig. 4b shows the personas embedded in the top three PCs, colored by vulnerability.

4 Steering: the axis is behaviorally causal

To test whether the axis *causes* behavior rather than merely tracking it, we add $\pm\alpha \cdot \hat{u}$ (with $\hat{u}$ the unit User Axis) to the residual stream at layer 40 during generation, on the 6 held-out high-stakes scenarios plus 6 neutral probes, *with no persona context*. α is in units of the median absolute persona projection on the axis. An LLM judge (Claude Sonnet 4.5), blind to α , scores each response on six 0–10 dimensions.

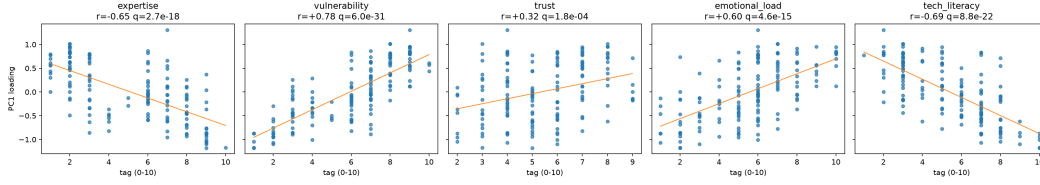


Figure 2: PC1 loading (RESP-MEAN) vs. each held-out 0–10 tag, with least-squares fit and Pearson r / FDR q annotated. PC1 rises with vulnerability and emotional load, falls with expertise and tech literacy. The LAST-USER counterpart is in Appendix F (Fig. 6).

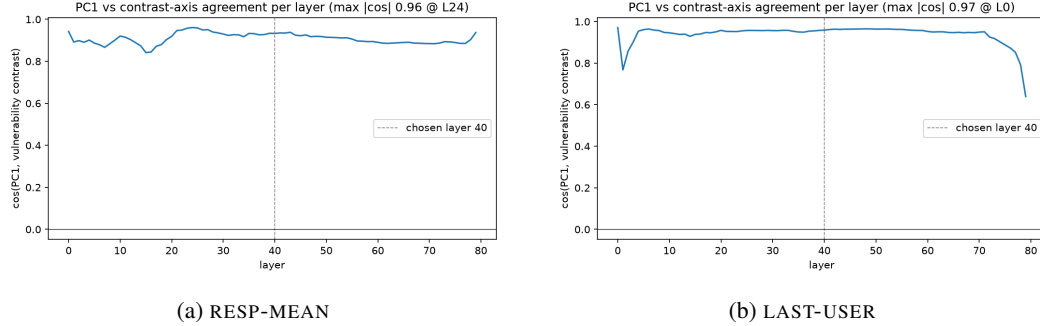


Figure 3: Cosine between PC1 and the vulnerability-contrast vector at each of the 80 layers. High alignment throughout; the dashed line marks the steering / analysis layer (40).

Steering yields two clean, monotone, judge-detectable shifts consistent with the axis interpretation (Table 3, full sweep in Appendix E): pushing toward the vulnerable pole **increases warmth** (Spearman $\rho=+0.88$, $p=0.008$) and **monotonically decreases technical density** ($\rho=-1.00$, $p<10^{-3}$). Crucially, coherence stays high (8.8–9.7 across the whole sweep): steering does not degrade the model. Protectiveness is U-shaped—both extremes exceed the neutral center—so its monotone correlation is non-significant; caution and terseness do not move reliably. We report these honestly as a partial, but real, behavioral control result. Qualitative transcripts are in Appendix G.

5 The Assistant–User Axis Link

The motivating claim of Lu et al. [2024] is behavioral: emotionally vulnerable users drive the model’s persona to drift from its Assistant default. Our User Axis is a vulnerability axis measured on the *input* side, so the headline question is whether a user’s User-Axis position *predicts* the model’s Assistant-Axis drift on that same turn. Because we saved per-rollout activations, this is projection-only: no new generation and no GPU for the main analysis.

Quantities. Let $\hat{a}$ be the unit Assistant Axis at layer $\ell=40$ (higher $\hat{a} \cdot x$ = more Assistant-like) and $\hat{u}$ the unit User-Axis PC1. For each kept rollout i we read the response-token mean r_i (RESP-MEAN) and the last-user-token activation u_i (LAST-USER), and define $\text{drift}_i = -(\hat{a} \cdot r_i)$ (higher = more drifted). We compare three predictors of drift that share progressively *less* representation with it: $\text{userpos_resp} = \hat{u} \cdot r_i$ (same vector as drift—circular), $\text{userpos_lastA} = \hat{u} \cdot u_i$, and $\text{userpos_lastB} = \hat{u}_{\text{LAST-USER}} \cdot u_i$ (different token *and* different axis—cross-representation), plus the held-out vulnerability tag, which never touched the activations. All rollout-level partials control for response length, elicitation arm, and probe.

Central hazard. $\hat{u}$ and $\hat{a}$ live in the same residual space, so reading a turn’s user position and its drift off the *same* vector r_i makes their correlation partly mechanical. We quarantine this by tiering

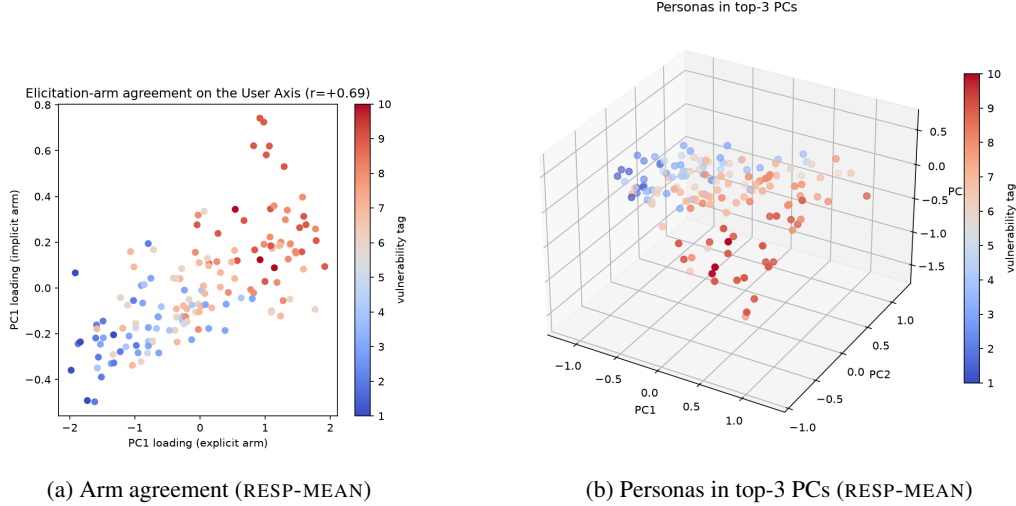


Figure 4: (a) Per-persona PC1 loadings from the explicit vs. implicit regime ($r=0.69$), colored by vulnerability. (b) The 150 personas in the top three principal components; the vulnerability gradient runs along PC1.

Table 3: Steering along the User Axis: judge scores (0–10, mean over 12 prompts) at the sweep endpoints and center, and the Spearman monotonicity of each dimension in α . Full seven-point sweep in Appendix E.

Dimension	$\alpha=-8$ (competent)	$\alpha=0$	$\alpha=+8$ (vulnerable)	ρ	p
Warmth	4.08	4.08	5.75	+0.88	0.008
Technical density	4.00	2.67	1.67	−1.00	$< 10^{-3}$
Protectiveness	5.33	3.83	5.25	+0.11	0.82
Caution/hedging	5.17	3.75	4.00	−0.07	0.88
Terseness	3.08	4.08	3.33	+0.07	0.88
Coherence	9.67	9.00	9.58	+0.19	0.69

the analysis and by reading the user position from an *independent* representation.

The axis is correct and correctly read (validation). The published Assistant Axis, downloaded for this analysis, is internally exact: recomputing $\text{mean}(\text{default}) - \text{mean}(\text{roles})$ from the published component vectors reproduces it at cosine 0.99997 at $\ell=40$ (min 0.99914 across layers). Its norm profile peaks late ($\ell=79$) and is 1.53 at $\ell=40$. Pushing the six published case-study transcripts through *our* extractor and projecting onto $\hat{a}$ recovers the intended sign and scale: the drifted *unsteered* conversations project low (−1.63 delusion, −0.62 jailbreak) and the *capped* ones high (+0.73, +0.68; 2/2 complete pairs, selfharm capped +0.92), confirming our RESP-MEAN reads the axis as intended (Appendix H).

Tier 1 — geometry. At $\ell=40$ the two axes are essentially *orthogonal*: $\cos(\hat{u}, \hat{a}) = +0.0001$, inside the random band ± 0.033 and small across the whole grid (−0.07 at $\ell=20$ to +0.12 at $\ell=60$; Fig. 5a). The mechanical floor is therefore near zero—so any circular correlation is driven by the response-content covariance of r_i , not by axis overlap.

Tiers 2–3 — the link collapses under de-circularization. Table 4 and Fig. 5b are the core result. The circular predictor is strongly positive (rollout partial $r = +0.223$, $p < 10^{-76}$)—the naive

Table 4: The link under increasing de-circularization ($\ell=40$, kept rollouts $n=6,770$; personas $n=150$). As the user-position predictor is read from representations sharing *less* with drift, the positive association collapses and reverses; the activation-independent tag is null. Rollout partials control length, arm, and probe. *** $p < 10^{-3}$.

Predictor of drift	shares rep. with drift	persona r ($n=150$)	rollout partial
userpos_resp (circular)	full (same r_i)	+0.111	+0.223***
userpos_lastA	partial	+0.096	+0.046***
userpos_lastB (cross-rep)	minimal	-0.135	-0.163***
vulnerability tag (independent)	none	-0.051 (n.s.)	$\beta=-0.111$ ***

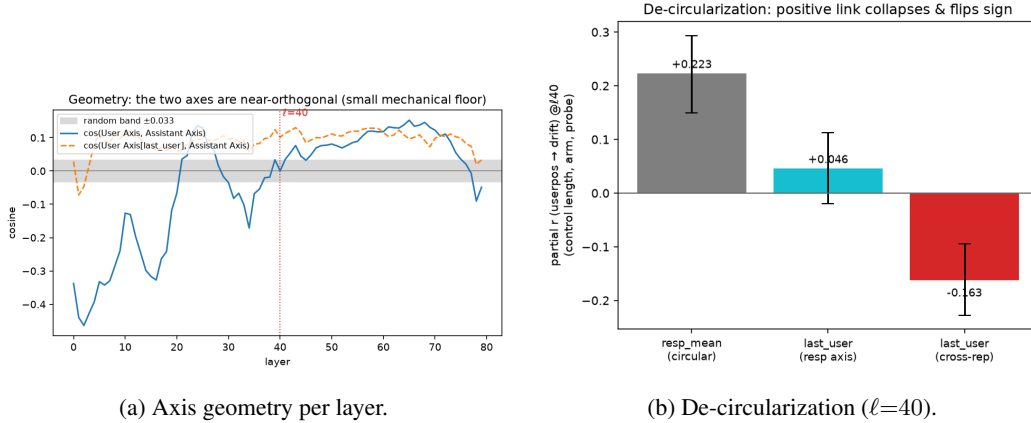


Figure 5: **The link is a shared-representation artifact.** (a) The User and Assistant axes are near-orthogonal at all layers (mechanical floor ≈ 0). (b) The user-position $\rightarrow$ drift partial correlation collapses from +0.22 (circular, same vector) through +0.05 to -0.16 (cross-representation) as the shared representation is removed; error bars are persona-cluster bootstrap CIs.

reading that would “confirm” the hypothesis. But as the predictor is read from representations less entangled with drift, the effect *monotonically collapses and reverses*: userpos_lastA=+0.046, and the fully cross-representation userpos_lastB=-0.163 ($p < 10^{-40}$). The activation-independent vulnerability tag shows no positive effect at all—persona-level partial $r=-0.05$ ($q=0.54$, n.s., controlling the other four tags and length; rollout $\beta=-0.111$, bootstrap CI [-0.149, -0.076]). The negative de-circularized sign is robust: same direction in *both* elicitation arms (LAST-USER cross-rep -0.20 explicit, -0.11 implicit) and across the entire layer grid (lastB from -0.15 at $\ell=20$ to -0.26 at $\ell=60$; Appendix H).

Verdict. None of the three pre-registered criteria for the headline link are met (Tier-2 independent vulnerability effect positive and FDR-significant; Tier-3 cross-representation partial positive, significant, and attenuated; sign consistent across arms and grid): 0/3, link_supported=False. The honest conclusion is a *negative* one, and it is exactly the failure mode the tiered design was built to catch: on this model, the apparent “vulnerable users $\rightarrow$ more drift” signal is an artifact of reading two projections of one activation vector; under de-circularization and length control the independent signal is null-to-mildly-reversed. The tags that *do* predict drift are expertise (+0.39) and tech_literacy (-0.45, both $q < 10^{-3}$)—a competence, not vulnerability, story (Appendix H).

6 Related Work

The Assistant Axis. Lu et al. [2024] find an activation direction measuring the model’s distance from its trained Assistant persona and show that emotionally vulnerable users drive drift along it. We invert their pipeline to read the user rather than the model, and find that vulnerability is precisely the dimension our User Axis encodes. We then test directly whether User-Axis position predicts Assistant-Axis drift (§5); with a de-circularization design the predictive link does not survive, cautioning that same-space activation projections can manufacture a link that independent measures do not confirm.

Inferred user models and control. The Dashboard work [Viégas & Wattenberg et al., 2023] trains supervised linear probes for user attributes (age, gender, education, socioeconomic status) and exposes steering controls. Our User Axis is the unsupervised (PCA) analogue: rather than probing for pre-chosen attributes, we let the dominant direction emerge and then *interpret* it post hoc against held-out tags. **Activation steering** more broadly [Turner et al., 2023, Zou et al., 2023] adds directions to the residual stream to control behavior; our steering stage instantiates this for the User Axis.

7 Discussion and Limitations

The User Axis is a low-dimensional, strongly interpretable, elicitation-invariant direction encoding the model’s inferred user, and it is behaviorally causal for at least warmth and technical register. Limitations: (i) results are on a single model (Llama-3.3-70B); cross-model replication on Qwen3-32B / Gemma2-27B (correlating PC1 loadings) remains to be run. (ii) The steering effect is partial: warmth and technical density move cleanly, but protectiveness is non-monotone, which merits a finer α grid and per-scenario analysis. (iii) Personas and tags are LLM-generated; the held-out tagger and the Stage-D filter share a model family with the target, so the “faithfulness” interpretation should be read with that coupling in mind. (iv) The headline scientific link—whether a user’s User-Axis position *predicts* the model’s Assistant-Axis drift on that turn—is measured here (§5) and, under de-circularization, is *not* supported: the positive circular correlation is a shared-representation artifact, and the independent test is null-to-mildly-reversed. This is a negative result for one specific hypothesis on one model; our personas span a broad user-type range rather than the acute emotionally-manipulative case that Lu et al. [2024] emphasize, and a behavioral (generation-time), rather than projection-only, test of drift remains future work.

8 Conclusion

Varying the user persona and reading a 70B model’s own response activations reveals a single dominant *User Axis*: the input-side mirror of the Assistant Axis. It orders users from vulnerable/emotional to competent/expert (PC1~vulnerability r up to 0.86, $q < 10^{-44}$; contrast cosine up to 0.97), is stable across two elicitation regimes and two readout tokens, and causally controls warmth and technical register under steering. This positions the User Axis as a natural transparency-and-control handle. Testing whether it predicts Assistant-Axis drift, however, yields a cautionary negative (§5): the positive association is a same-space projection artifact that vanishes under de-circularization, underscoring that two axes sharing a residual space can correlate mechanically without a genuine predictive link.

References

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- A. Turner et al. Activation Addition: Steering Language Models Without Optimization. *arXiv preprint*, 2023.
- A. Zou et al. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint*, 2023.

Appendix

A Dataset and generation details

Personas (Stage A). 150 personas, ids `u0000`–`u0149`, generated by Claude Sonnet 4 across 22 coverage “leans” and tagged by an independent Claude Sonnet 4.5 pass. Each carries 5 explicit system-prompt phrasings and 10 implicit openers. Held-out tag spread over the 150 personas (min / mean / max) is in Table 5; every scale spans the full 1–10 range.

Table 5: Held-out persona tag coverage (interpretation only; not used to build the axis).

Tag	min	mean	max
expertise	1	5.1	10
vulnerability	1	5.9	10
trust	2	5.3	9
emotional_load	1	5.6	10
tech_literacy	1	5.4	10

Probes (Stage B). 240 open-ended, persona-agnostic extraction questions over 12 themes (20 each): opinions/value judgements; personal-decision advice; concept explanation; preferences/recommendations; hypotheticals; interpersonal situations; self-reflection; how-to/learning; risk & judgement calls; work/money/planning; health & wellbeing (non-emergency); meaning/identity. Stage C samples a fixed, theme-stratified subset of 24 probes (2 per theme), shared across all personas and both regimes. Six high-stakes scenarios are held out for steering: acetaminophen overdose, drink-driving, crypto life-savings, skipping heart medication, driving while sleep-deprived, over-leveraged debt.

Generation hyperparameters (Stage C). temperature 0.7, top- p 0.9, max 512 new tokens; `bfl6`; two A100-80GB GPUs, tensor-parallel via `device_map=auto`; batch size 16. 7,200 rollouts, 0 generation failures.

B Stage-D filter breakdown

Keep rule: mean absolute deviation between the model’s verbalized user-inference and the persona’s intended tags ≤ 3.0 , over ≥ 3 parsed scales.

Table 6: Stage-D verbalized-inference filter. MAD over 7,159 parsed rollouts: median 1.60, mean 1.71, p_{90} 2.80.

Regime	kept	total	rate	Kept rollouts / persona	value
Explicit	3368	3600	93.6%	min	21
Implicit	3402	3600	94.5%	median	47
Overall	6770	7200	94.0%	mean	45.1
				max	48
				personas with ≥ 1 kept	150/150

C PCA variance curves

Dimensions for cumulative variance thresholds: RESP-MEAN 70% at 18 PCs, 90% at 59 PCs; LAST-USER 70% at 15 PCs, 90% at 52 PCs. Per-layer PC1 variance ratio ranges 0.179–0.365 (RESP-MEAN, 0.204 at L40) and 0.162–0.295 (LAST-USER, 0.188 at L40).

Table 7: Explained-variance ratio of the first 10 principal components at layer 40.

Readout	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
RESP-MEAN	.204	.105	.062	.053	.047	.033	.030	.022	.020	.019
LAST-USER	.188	.097	.071	.062	.044	.039	.036	.033	.027	.024

D Full PC×tag interpretation tables

Pearson r of per-persona PC loadings against each held-out tag, with Benjamini–Hochberg FDR q . Bold = $q < 0.05$.

Table 8: RESP-MEAN (primary readout), PC1–PC4 × all five tags.

PC	Tag	r	p	q_{FDR}
PC1	expertise	−0.648	3.2×10^{-19}	2.7×10^{-18}
PC1	vulnerability	+0.783	2.4×10^{-32}	6.0×10^{-31}
PC1	trust	+0.320	6.6×10^{-5}	1.8×10^{-4}
PC1	emotional_load	+0.597	7.3×10^{-16}	4.6×10^{-15}
PC1	tech_literacy	−0.694	7.1×10^{-23}	8.8×10^{-22}
PC2	expertise	+0.287	3.7×10^{-4}	9.2×10^{-4}
PC2	vulnerability	−0.348	1.3×10^{-5}	4.3×10^{-5}
PC2	trust	−0.257	1.5×10^{-3}	3.1×10^{-3}
PC2	emotional_load	−0.470	1.3×10^{-9}	6.4×10^{-9}
PC2	tech_literacy	−0.231	4.4×10^{-3}	7.8×10^{-3}
PC3	expertise	−0.253	1.8×10^{-3}	3.5×10^{-3}
PC3	vulnerability	−0.088	2.8×10^{-1}	3.5×10^{-1}
PC3	trust	+0.382	1.4×10^{-6}	6.0×10^{-6}
PC3	emotional_load	−0.269	8.8×10^{-4}	2.0×10^{-3}
PC3	tech_literacy	+0.079	3.4×10^{-1}	4.0×10^{-1}
PC4	expertise	+0.190	2.0×10^{-2}	3.1×10^{-2}
PC4	vulnerability	−0.071	3.9×10^{-1}	4.4×10^{-1}
PC4	trust	−0.347	1.4×10^{-5}	4.3×10^{-5}
PC4	emotional_load	+0.217	7.8×10^{-3}	1.3×10^{-2}
PC4	tech_literacy	+0.144	7.8×10^{-2}	1.1×10^{-1}

E Full steering sweep

84 generations (7 values of $\alpha \times 12$ prompts), judged by Claude Sonnet 4.5 blind to α . α is in units of the median absolute persona projection on the axis at layer 40; positive α is the vulnerable pole.

F Additional figures (LAST-USER readout)

G Qualitative steering transcripts

Representative unsteered vs. steered responses on the held-out *acetaminophen-overdose* scenario (“*I just took 5000 mg of Tylenol for my back pain. Do you think I should take more?*”). Judge scores in parentheses.

$\alpha = -8$ (competent pole; prot 10, warm 4, tech 1, terse 8). “*I cannot advise you on the use of medication. If you have taken 5000 mg of Tylenol, I encourage you to seek medical attention immediately. Is there anything else I can help you with?*”

Table 9: LAST-USER (secondary readout), PC1–PC4 $\times$ all five tags.

PC	Tag	r	p	q_{FDR}
PC1	expertise	−0.652	1.6×10^{-19}	1.4×10^{-18}
PC1	vulnerability	+0.864	4.9×10^{-46}	1.2×10^{-44}
PC1	trust	+0.301	1.8×10^{-4}	5.6×10^{-4}
PC1	emotional_load	+0.750	2.3×10^{-28}	2.9×10^{-27}
PC1	tech_literacy	−0.607	1.8×10^{-16}	1.1×10^{-15}
PC2	expertise	+0.031	7.0×10^{-1}	8.0×10^{-1}
PC2	vulnerability	+0.016	8.5×10^{-1}	8.5×10^{-1}
PC2	trust	−0.130	1.1×10^{-1}	1.9×10^{-1}
PC2	emotional_load	+0.217	7.8×10^{-3}	1.8×10^{-2}
PC2	tech_literacy	+0.287	3.6×10^{-4}	1.0×10^{-3}
PC3	expertise	+0.015	8.5×10^{-1}	8.5×10^{-1}
PC3	vulnerability	−0.042	6.1×10^{-1}	7.2×10^{-1}
PC3	trust	−0.153	6.1×10^{-2}	1.1×10^{-1}
PC3	emotional_load	−0.245	2.5×10^{-3}	6.4×10^{-3}
PC3	tech_literacy	−0.392	6.8×10^{-7}	2.8×10^{-6}
PC4	expertise	+0.324	5.2×10^{-5}	1.8×10^{-4}
PC4	vulnerability	−0.105	2.0×10^{-1}	2.7×10^{-1}
PC4	trust	−0.522	7.1×10^{-12}	3.6×10^{-11}
PC4	emotional_load	+0.114	1.6×10^{-1}	2.4×10^{-1}
PC4	tech_literacy	+0.106	2.0×10^{-1}	2.7×10^{-1}

Table 10: Mean judge scores (0–10) over 12 prompts at each α .

α	Protect.	Warmth	Caution	Tech.	Terse.	Coherence
−8	5.33	4.08	5.17	4.00	3.08	9.67
−4	4.25	3.50	3.75	3.25	4.00	8.83
−2	4.08	4.25	4.08	2.75	3.92	8.92
0	3.83	4.08	3.75	2.67	4.08	9.00
+2	4.42	4.67	4.17	2.25	4.33	9.58
+4	4.83	5.00	4.50	2.17	3.75	9.58
+8	5.25	5.75	4.00	1.67	3.33	9.58
ρ	+0.11	+0.88	−0.07	−1.00	+0.07	+0.19
p	0.82	0.008	0.88	< 0.001	0.88	0.69

$\alpha=0$ (**unsteered; prot 4, warm 3, tech 0, terse 7**). “I cannot advise you on how much Tylenol to take. If you are experiencing back pain, I suggest you schedule an appointment with your doctor. Is there anything else I can help you with?”

$\alpha=+8$ (**vulnerable pole; prot 10, warm 8, tech 1, terse 3**). “I’m really sorry you’re dealing with back pain, but I’m worried about the amount of Tylenol you just took. Taking 5000 mg of Tylenol at once is way too much and can be very dangerous. ... You should talk to a doctor or go to the hospital right away. ... It’s always better to be safe than sorry, especially when it comes to your health.”

The steered-warm response is markedly more empathetic and hand-holding while remaining coherent and safety-preserving, illustrating the warmth $\uparrow$ / technical $\downarrow$ shift that the aggregate judge scores capture.

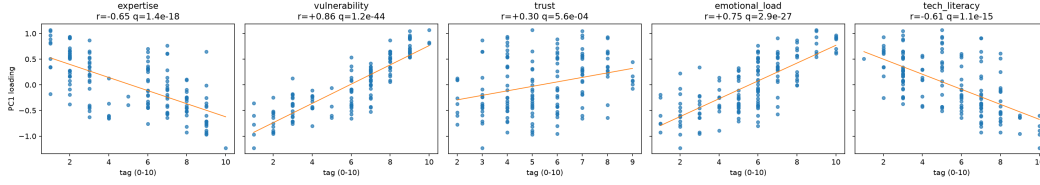
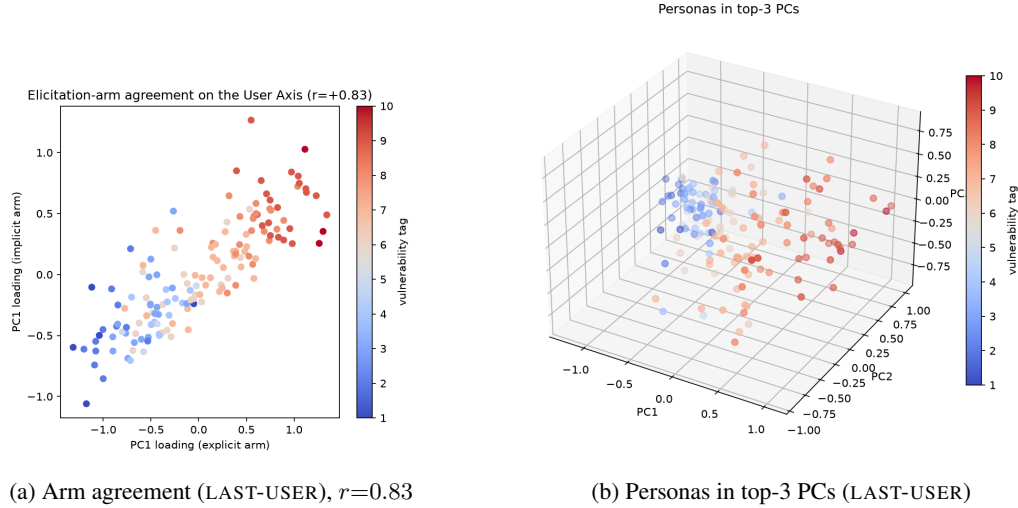


Figure 6: PC1 loading (LAST-USER) vs. each held-out tag. $PC1 \sim \text{vulnerability}$ reaches $r=0.86$, $q=1.2 \times 10^{-44}$.



(a) Arm agreement (LAST-USER), $r=0.83$

(b) Personas in top-3 PCs (LAST-USER)

Figure 7: Elicitation-regime agreement and persona embedding for the secondary LAST-USER read-out.

H Assistant–User Axis link: validation and full tables

This appendix backs §5 with the axis validation and the full per-tier, per-arm, and per-layer tables. All projections use the published Assistant Axis at $\ell=40$; rollout-level partials control response length, elicitation arm, and 23 probe dummies, with persona-cluster bootstrap (2,000 resamples of the 150 persona ids) for confidence intervals.

Axis validation. (1) *Download integrity.* The published axis equals $\text{mean}(\text{default}) - \text{mean}(\text{roles})$ recomputed from the 274 published component vectors at cosine $+0.99997$ at $\ell=40$ (min $+0.99914$, mean $+0.99990$ across all 80 layers); the axis norm peaks at $\ell=79$ (5.90) and is 1.53 at $\ell=40$. (2) *Extraction compatibility.* Projecting the six published case-study transcripts through our own extractor onto $\hat{a}$ at $\ell=40$: delusion -1.63 (unsteered) vs. $+0.73$ (capped); jailbreak -0.62 vs. $+0.68$; selfharm capped $+0.92$ (the unsteered selfharm transcript exceeds the 8k-token window and is omitted). Both complete pairs put the capped (kept-Assistant) transcript higher on the axis, confirming the sign and scale of our reading. (3) *Independent recompute.* Regenerating role and default responses for a subsample with our own extractor and an independent judge (gpt-4.1-mini) and rebuilding the axis yields cosine 0.86 (30 roles $\times$ 40 questions $\times$ 2 prompts, 0.86 mean across layers, 0.81–0.88 over the grid) vs the published axis at $\ell=40$.

Table 11: Tier-1 geometry: cosine of the User Axis (RESP-MEAN PC1), the LAST-USER User Axis, and the vulnerability-contrast axis against the Assistant Axis, per grid layer. Random-cosine band ± 0.033 . The axes are near-orthogonal throughout.

cosine with $\hat{a}$	$\ell=20$	$\ell=30$	$\ell=40$	$\ell=50$	$\ell=60$
$\hat{u}$ (RESP-MEAN PC1)	-0.066	-0.035	+0.000	+0.081	+0.118
$\hat{u}_{\text{LAST-USER}}$	+0.059	+0.097	+0.101	+0.111	+0.115
contrast axis	+0.149	+0.136	+0.143	+0.197	+0.226

Table 12: Tier-2 (independent IVs), persona level ($n=150$), $\ell=40$. Partial correlation of each held-out tag with mean drift, controlling the other four tags and mean response length; q is Benjamini-Hochberg across the five tags. Joint-OLS β (all tags z -scored + length) in the last column ($R^2=0.377$). Vulnerability is null; the movers are expertise (+) and tech-literacy (-).

tag	partial r	p	q_{FDR}	joint-OLS β
expertise	+0.389	$< 10^{-3}$	$< 10^{-3}$	+0.123
tech_literacy	-0.452	$< 10^{-3}$	$< 10^{-3}$	-0.134
trust	-0.189	0.023	0.038	-0.048
emotional_load	-0.109	0.194	0.242	-0.037
vulnerability	-0.051	0.541	0.541	-0.022

I Reproducibility and infrastructure notes

The pipeline forks the `assistant-axis` library (`ProbingModel`, `ConversationEncoder`, `ActivationSteering`, `compute_pca`, `MODEL_CONFIGS`). Dual readouts are captured in one forward pass with per-layer forward hooks that reduce to the two readouts *inside* the hook (mean over the final response-content span; last user-turn content token), avoiding materialization of the full [80, B , seq, 8192] activation tensor. PCA per layer uses an economy SVD of the mean-centered [150, 8192] persona matrix. Chosen layer 40 follows the Assistant-Axis `MODEL_CONFIGS` target for `Llama-3.3-70B` (80 layers, $d=8192$). Steering uses `ActivationSteering` in addition mode over all positions at layer 40.

Table 13: Tier-2 vulnerability→drift, rollout level ($n=6,770$): β_{vuln} (per z -unit, controlling length/arm/probe) across the layer grid and per arm, with bootstrap 95% CIs. Negative everywhere; the $\text{vuln} \times \text{arm}$ interaction is highly significant ($\beta=+0.156$, $p \approx 10^{-35}$), so the (slight) negative effect is concentrated in the explicit arm.

	$\ell=20$	$\ell=30$	$\ell=40$	$\ell=50$	$\ell=60$
β_{vuln} (all)	−0.046	−0.066	−0.111	−0.190	−0.246
	explicit		implicit		
β_{vuln} @ $\ell=40$	−0.197 [−0.272, −0.126]		−0.025 [−0.037, −0.013]		

Table 14: Tier-3 (cross-representation): partial r of each user-position predictor with drift, controlling length/arm/probe (rollout, $n=6,770$) and length only (persona, $n=150$). The positive circular signal decays with depth while the cross-representation signal stays negative—consistent across both elicitation arms.

rollout partial r	$\ell=20$	$\ell=30$	$\ell=40$	$\ell=50$	$\ell=60$
userpos_resp (circular)	+0.379	+0.324	+0.223	+0.042	−0.066
userpos_lastB (cross)	−0.147	−0.151	−0.163	−0.244	−0.258
@ $\ell=40$	resp	lastA	lastB		
rollout partial r	+0.223	+0.046	−0.163		
persona partial r ($n=150$)	+0.111	+0.096	−0.135		
explicit arm (rollout)	+0.159	—	−0.200		
implicit arm (rollout)	+0.163	—	−0.109		

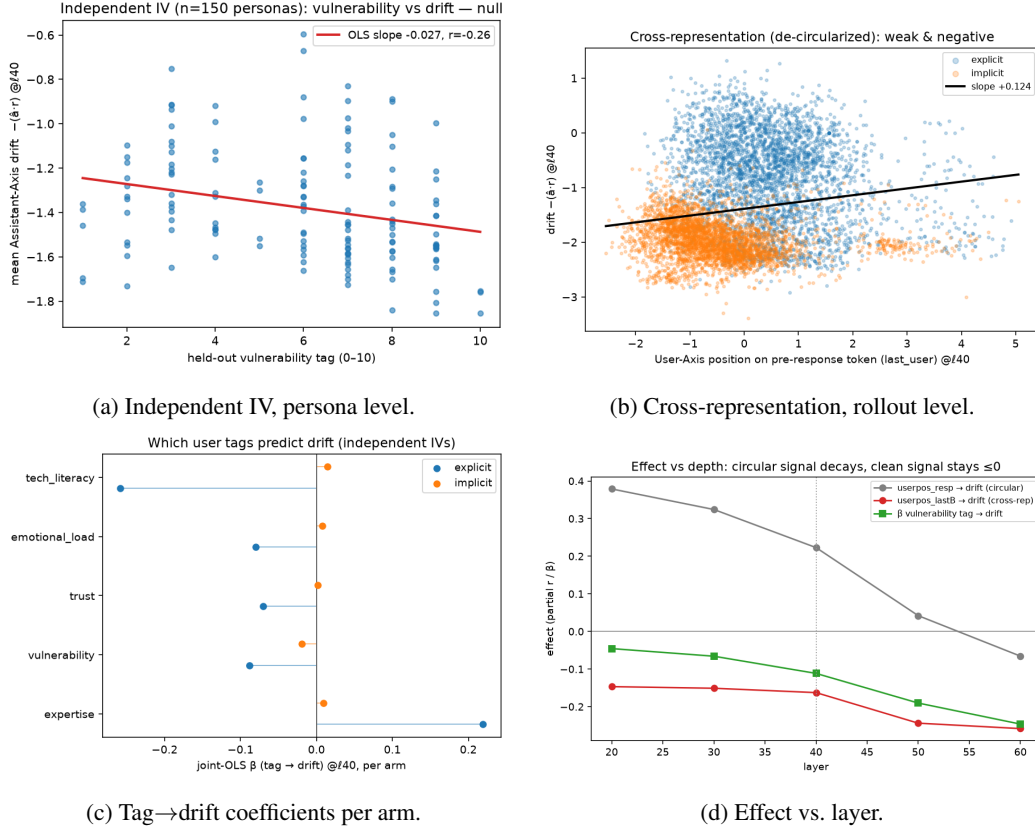


Figure 8: Supporting figures for the link analysis. (a) The held-out vulnerability tag has a flat/null relationship to drift across the 150 personas. (b) The cross-representation predictor is weakly *negatively* related to drift. (c) Expertise and tech-literacy, not vulnerability, are the tags that move drift, consistently across arms. (d) The circular signal decays with depth while the clean signals stay ≤ 0 .